Term 7- Sept 2025

Nanoelectronics and Technology (01.119/99.503)-Week 10 Class 1

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18-Nov- 2025

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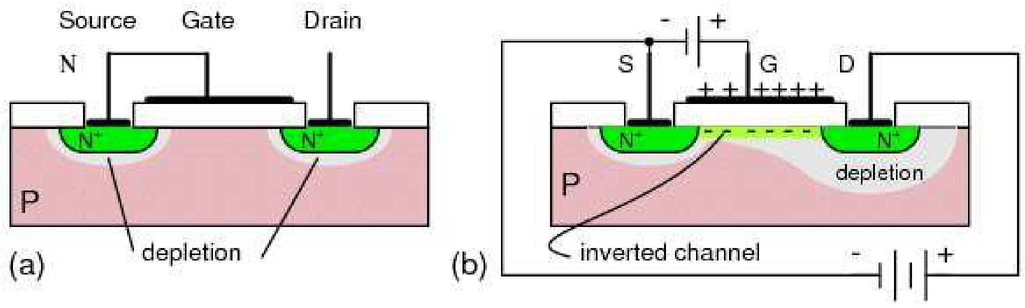
Outline

- Nanoscale characterization
  - Scanning tunneling microscopy (STM)
  - Atomic force microscopy (AFM)

Device Level Characterization

Conventional Device level Electrical Analysis

- I-V Characterization, I-t Characterization
- C-V characterization
- Breakdown analysis at the device level



- Drawback: It is difficult characterize the materials and device locally at the nanometer scale
- Overall average effect

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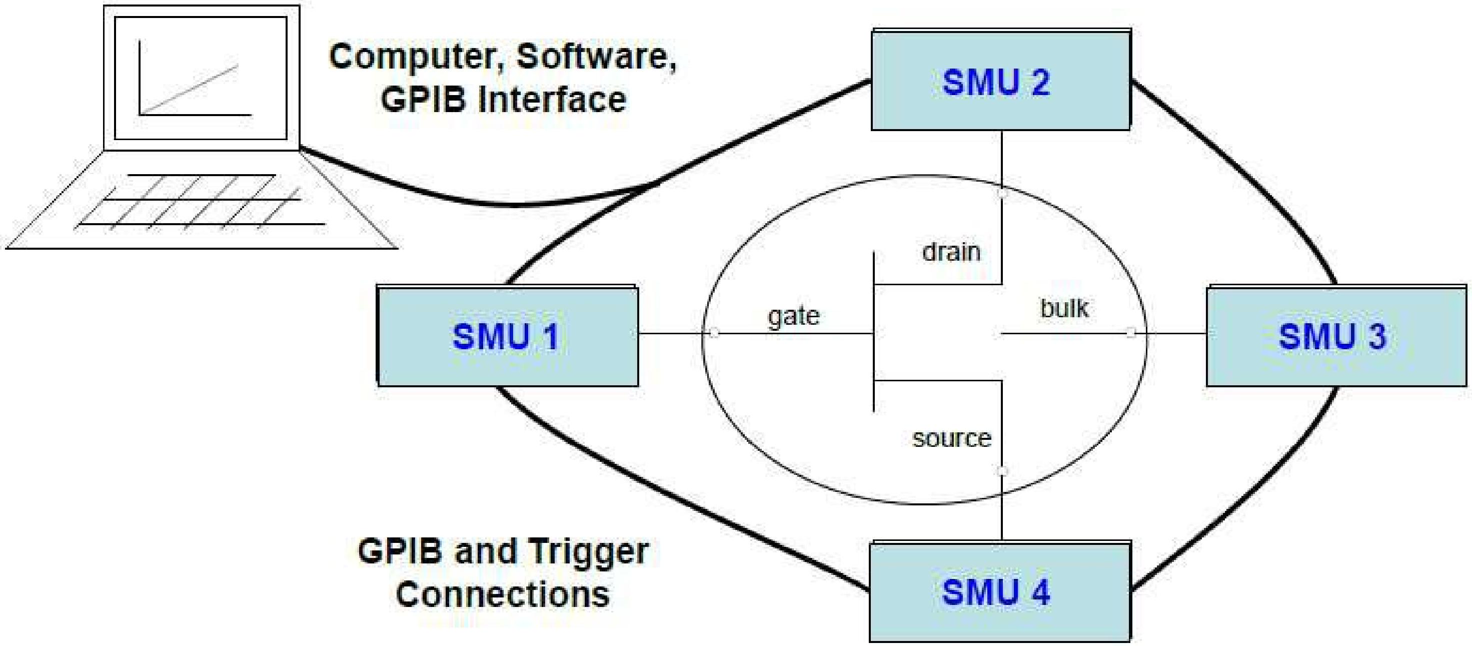
Probe station Electrical Characterization

Automating Multiple SMUs

Automating and testing of multi-terminal devices requires the user to create a highly complex system involving programming of multiple instruments, precise triggering, graphics, analysis, etc.



Probe station Electrical Characterization-MOSFET

- **Output Characteristics (I_D vs. V_{DS}):** Measures the drain current (I_D) as a function of the drain-source voltage (V_{DS}) for different constant gate-source voltages (V_{GS}). These curves show the device's behavior in the linear and saturation regions.

- **Transfer Characteristics (I_D vs. V_{GS}):** Measures the drain current (I_D) as a function of the gate-source voltage (V_{GS}) at a constant V_{DS}. This is used to determine the **threshold voltage (V_{th})**, the minimum gate voltage required to create a conductive channel.

- **Leakage Currents:** Characterization includes measuring parasitic currents like gate leakage current (I_{GSS}) and drain cut-off current (I_{DSS}) to assess device quality and off-state performance.

- **Breakdown Voltage ($V_{(BR)}$):** Determines the maximum voltage the device can block between the drain and source before experiencing breakdown.

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Nanoscale Characterization

Nanoscale level characterization presents unique physical, electrical and chemical properties compared to their respective particles at higher scales (device level).

- Scanning Probe Microscopy (SPM)
  - Scanning Tunneling Microscopy (STM)
  - Atomic Force Microscopy (AFM)
- Electron Microscopy (TEM)
  - Transmission Electron Microscopy (TEM)
  - Scanning Electron Microscopy (SEM)
- Spectroscopic Techniques
  - X-ray Photoelectron Spectroscopy (XPS)
  - Energy-dispersive X-ray spectroscopy (EDX)

Nanoscale Characterization

Nanoscale level characterization presents unique physical, electrical and chemical properties compared to their respective particles at higher scales (device level).

- **Small structural variations (at nanoscale) create large electrical variations**
  - Missing Si atoms → threshold voltage shifts
  - Small roughness → mobility degradation
- We need **high-resolution techniques** to measure:
  - interface roughness
  - gate oxide thickness
  - line-edge roughness
- Most reliability and failures in advanced CMOS are **nanometer-scale phenomena**
- Without nanoscale characterization, nanoelectronics cannot be understood, optimized, or scaled effectively.

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Ultra-High Vacuum (UHV) condition

- UHV conditions are created by pumping the gas out of a UHV chamber.

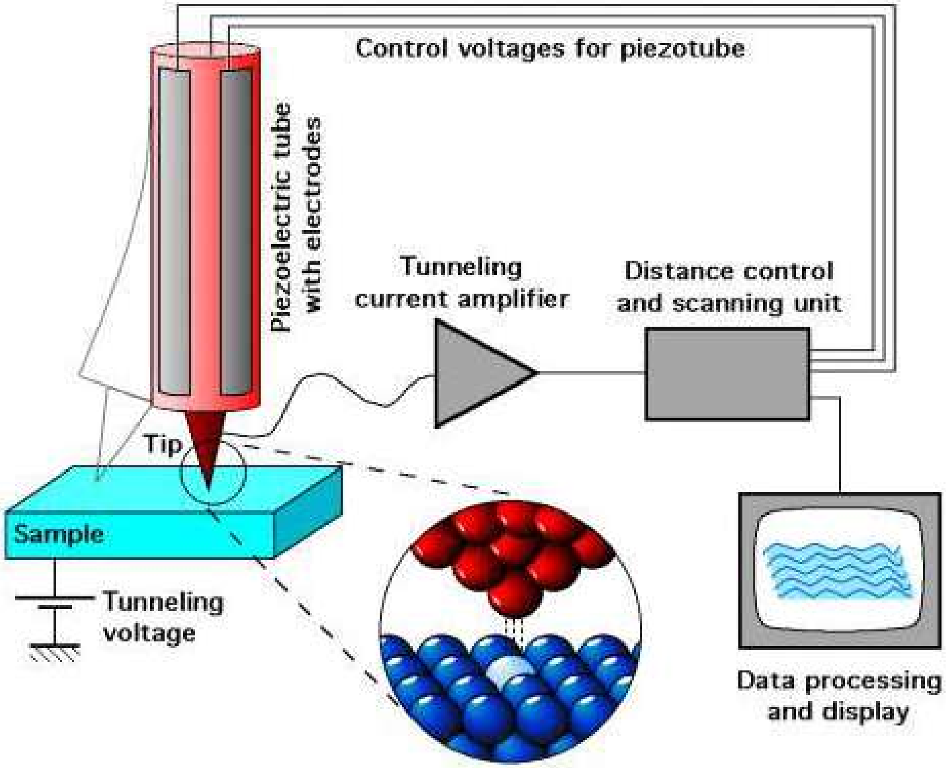
- As the vacuum is the absence of material, what is in fact measured is the pressure of residual gas in the chamber. Vacuum represents particle density lower than the atmospheric pressure.

- UHV conditions are integral to scientific research and Surface science experiments often require a chemically clean sample surface with the absence of any unwanted adsorbates.

https://www.orsayphysics.com/what-is-uhv

Scanning Tunneling Microscopy

- STM-tunneling of electrons from sample to tip and vice versa through vacuum
- UHV-conditions, Tip- Pt and W



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STM Applications

- Surface structure analysis compared to Bulk material- Topography at the atomic level/nanometer scale
- Surface structure analysis applicable to wide variety of samples
- Spectroscopy of local area of the samples- Extract local electrical characteristics of the materials

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Key Challenges in STM

- Sample Preparation-High Surface Quality
- **Material Compatibility:** Some materials, such as insulators, are difficult to study with STM due to their lack of conductivity. Although variations like scanning tunneling spectroscopy (STS) and atomic force microscopy (AFM) can address this, conducting materials are preferred.
- **Handling Sensitive Samples**: Some materials, particularly biological samples or organic molecules, can be damaged under the high-energy electron beam used in STM, requiring careful handling or cryogenic conditions.
- **Tip-Sample Interaction (Tip Stability):** The sharpness and cleanliness of the STM tip are crucial for high-resolution imaging. Over time, the tip can become contaminated or blunt, leading to inaccurate results. Maintaining a sharp, clean tip in operational conditions is challenging, and sometimes requires periodic tip renewal or treatment (e.g., scratching/heating to restore sharpness).
- **Environmental Sensitivity** (Vibration Sensitivity) : STM requires an extremely stable environment, as even minor vibrations can interfere with measurements. Laboratories typically require vibration isolation systems to minimize this issue.

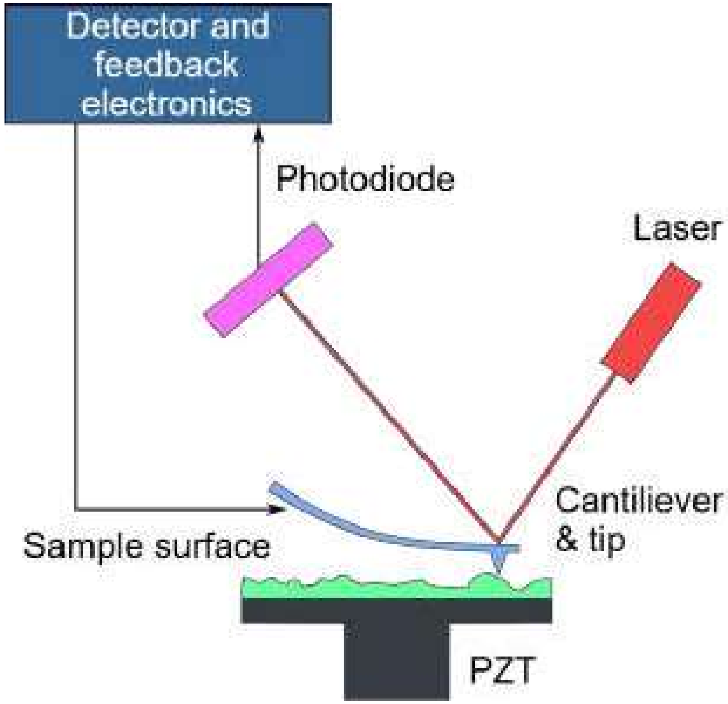
Atomic Force Microscopy

- Atomic Force Microscopy (AFM) is a **high-resolution scanning probe technique** capable of resolving surface features down to **sub-nanometer** scale. It works by **measuring the interaction forces** between a sharp probe (tip) and the sample surface.

Basic Components

AFM consists of:

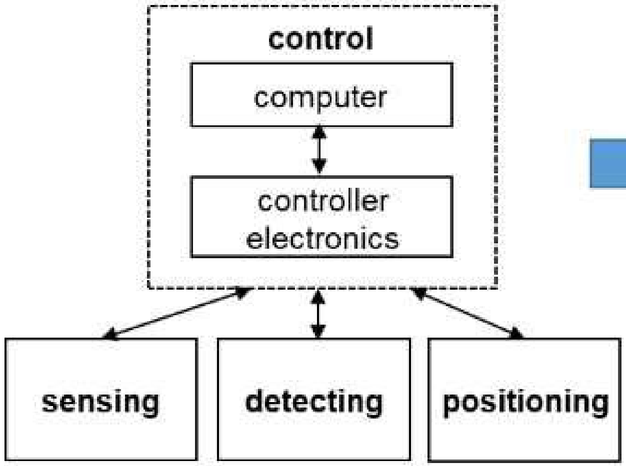
1. **Cantilever with a sharp tip**
   - Tip radius: **5–10 nm**
   - Cantilever spring constant: **0.01–100 N/m**
2. **Piezoelectric scanner**
   - Moves the sample (or tip) in **x–y–z** directions with **Å-level precision.**
3. **Laser and Position-Sensitive Photodiode (PSPD)**
   - To detect cantilever deflection.
4. **Feedback control electronics**
   - Keeps the interaction force constant.

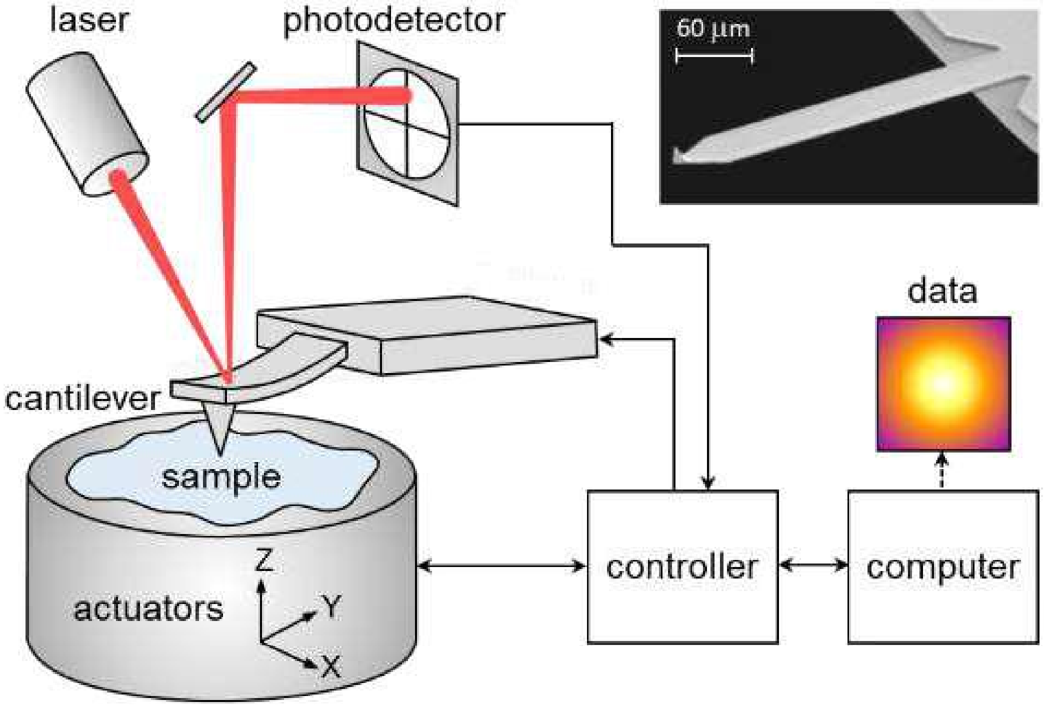


https://en.wikipedia.org/wiki/Atomic_force_microscopy

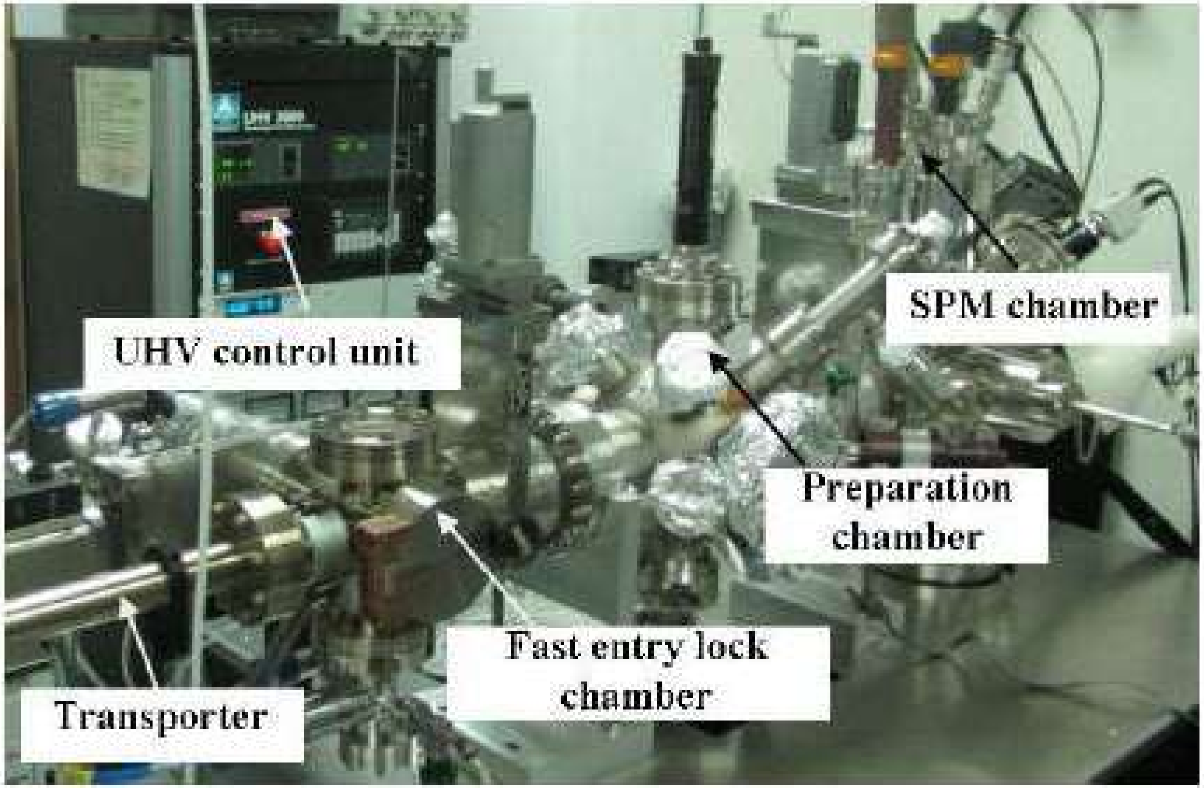
Atomic Force Microscopy

- Contact AFM: To acquire both topography and current maps
- Non-contact mode AFM: Topography
- Tapping mode-AFM: Topography



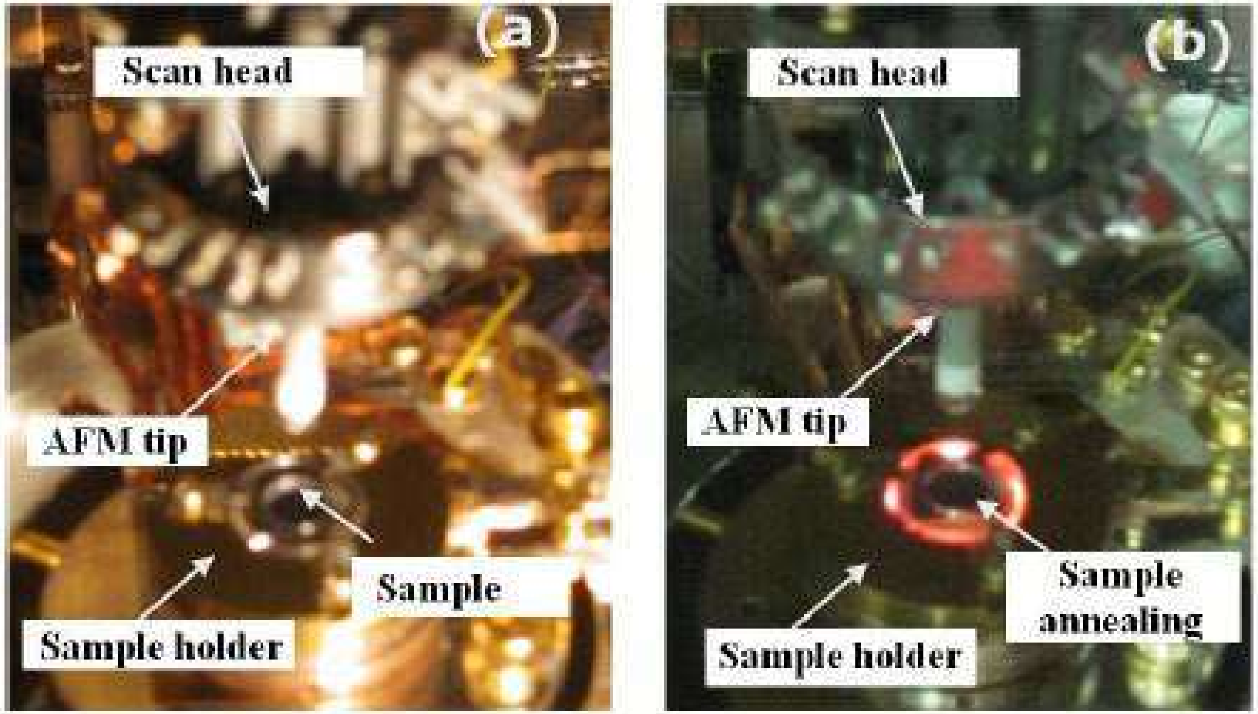


Atomic Force Microscopy



The UHV-SPM (RHK) system consisting of a fast entry lock chamber, a preparation chamber and SPM chamber.

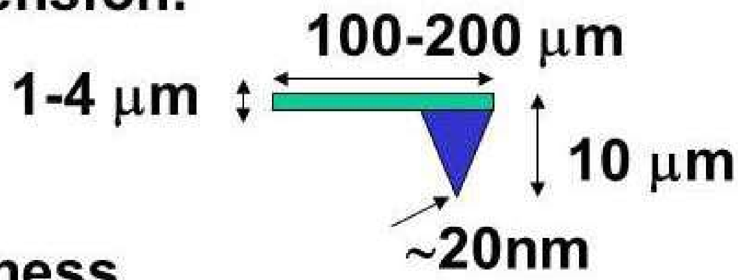
Atomic Force Microscopy



(a) Image of the scan head assembly and sample holder, and (b) during sample annealing.

Atomic Force Microscopy

- **Material:** **Silicon, Pt-Ir, Diamond, etc**
- **Dimension:**



- **Stiffness**
  - **soft: contact mode**
  - **stiff: vibrating (dynamic force) mode**
- **Spring constant (k): 0.1-10N/m**
- **Resonance frequency: 10~100kHz**

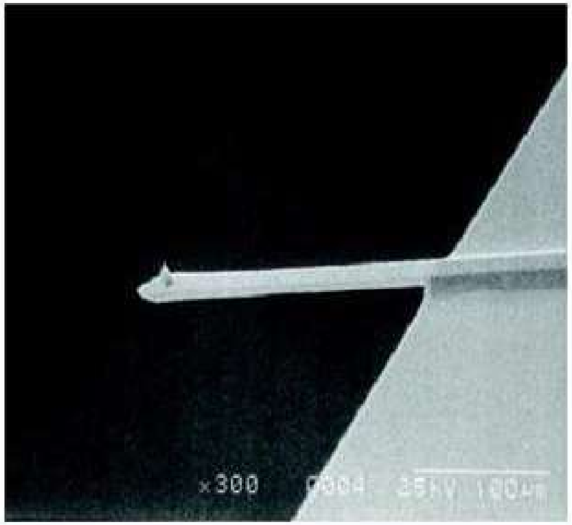


Figure 3. SEM image of an integrated single crystal silicon cantilever and tip which has an end radius of 2 to 10nm. Tips for AFM are typically made of silicon or silicon nitride. Bar=100µm.

Atomic Force Microscopy

Force

How are Forces Measured?

The probe is placed on the end of a cantilever (which one can think of as a spring). The amount of force between the probe and sample is dependant on the *spring constant* (stiffness) of the cantilever and the distance between the probe and the sample surface. This force can be described using Hooke's Law:

$$F=-k\cdot x$$

F = Force
k = spring constant
x = cantilever deflection

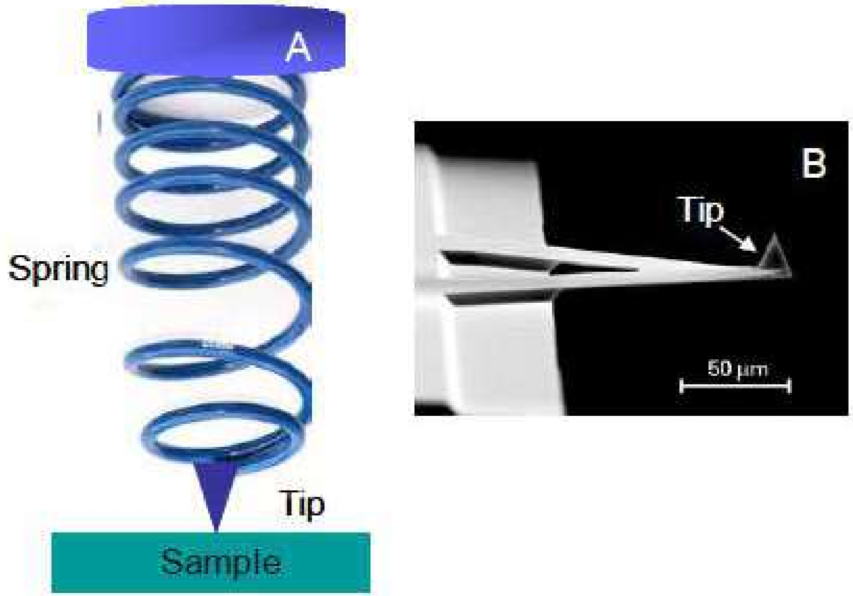


Figure 1. a) Spring depiction of cantilever b) SEM image of triangular SPM cantilever with probe (tip). (Image from MikroMasch)

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Atomic Force Microscopy-Tip properties

The **spring constant (k) of the AFM tip's cantilever** plays a crucial role in both **contact** and **non-contact** modes, affecting sensitivity, resolution, and sample interaction.

1. Contact Mode and Spring Constant

- In **contact mode**, the tip remains in direct contact with the sample surface.
- The force applied by the tip follows **Hooke's Law**:
- $F = k \cdot d$ where **F** is the force, **k** is the spring constant of the cantilever, and **d** is the deflection.

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- A **higher spring constant (stiffer cantilever)** is needed to avoid unwanted bending when scanning **hard materials** like metals and semiconductors.

Atomic Force Microscopy-Tip properties

2. Non-Contact Mode and Spring Constant

- In **non-contact mode**, the tip hovers just above the surface, detecting long-range forces (van der Waals, electrostatic, etc.) without touching.
- The cantilever oscillates at its **resonance frequency**, and **small force variations** shift this frequency.

  - A **low spring constant** is preferred, as a more flexible cantilever is more sensitive to small force changes.
  - Useful for **soft biological samples** to minimize sample damage.

Atomic Force Microscopy-Tip properties

Key Considerations:

Typical spring constant ranges:

Contact mode: k≈1−50 N/m (depends on material hardness).

Non-contact mode: k≈1−10 N/m (enough stiffness to maintain oscillation).

- In **contact mode**, the spring constant determines the applied force and potential sample damage.
- In **non-contact mode**, it affects oscillation stability and sensitivity to long-range forces.

Choosing the right spring constant is critical for **minimizing damage, maximizing resolution, and obtaining accurate measurements**.

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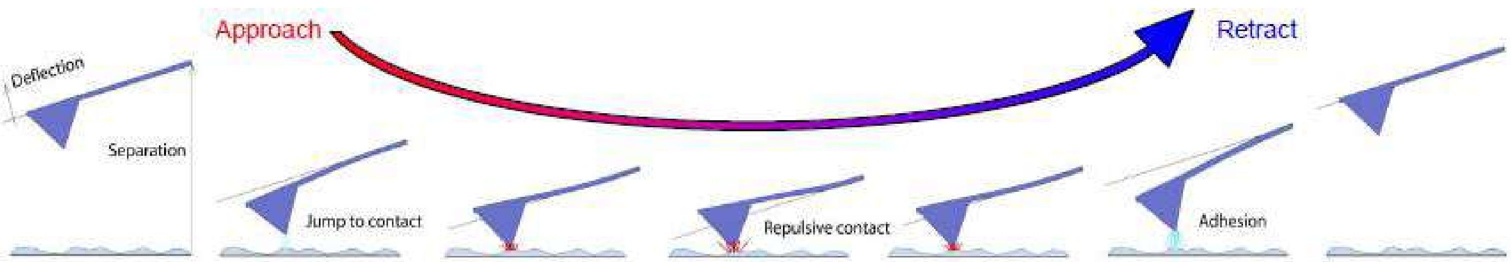
Atomic Force Microscopy

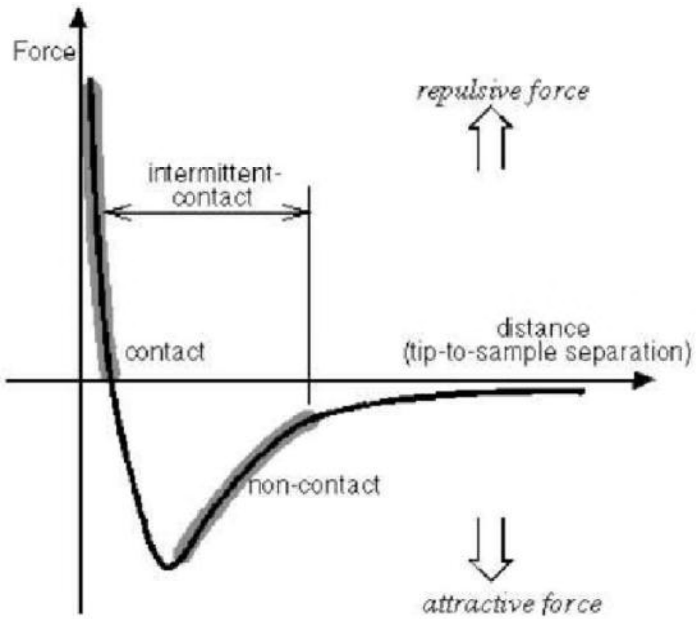
Force –distance curve

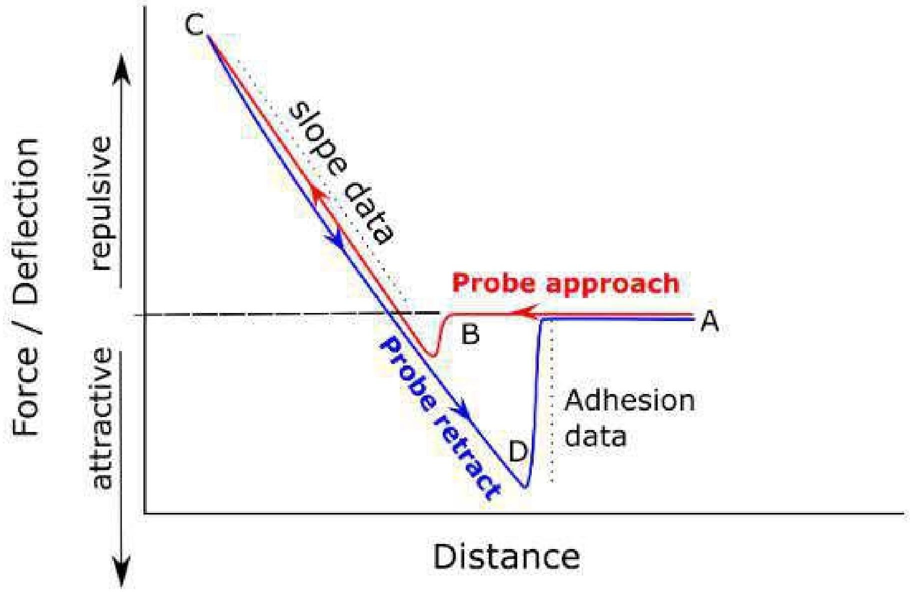
Atomic Force Microscopy

Force –distance curve



Schematic diagram of the vertical tip movement during the approach and retract parts of a force spectroscopy experiment.





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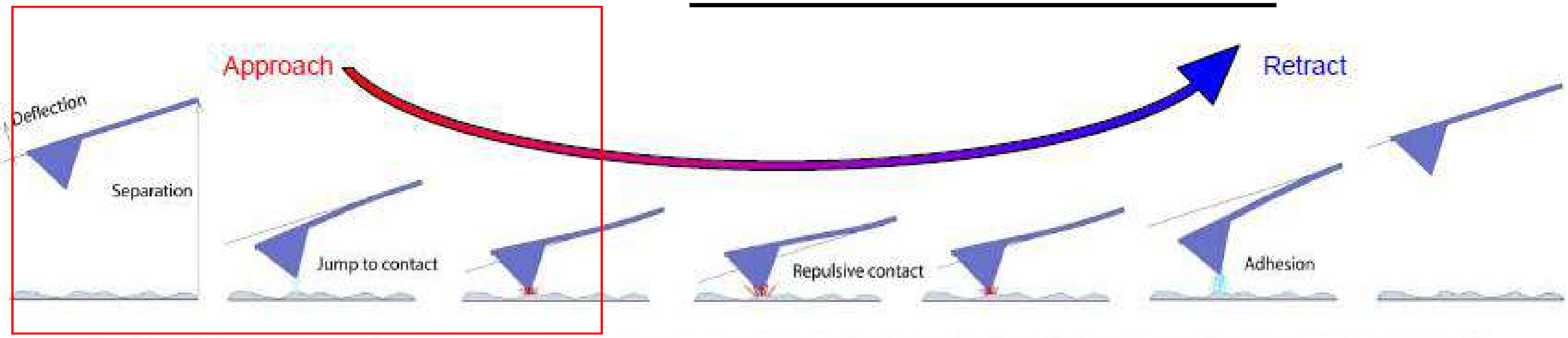
Atomic Force Microscopy

Force –distance curve



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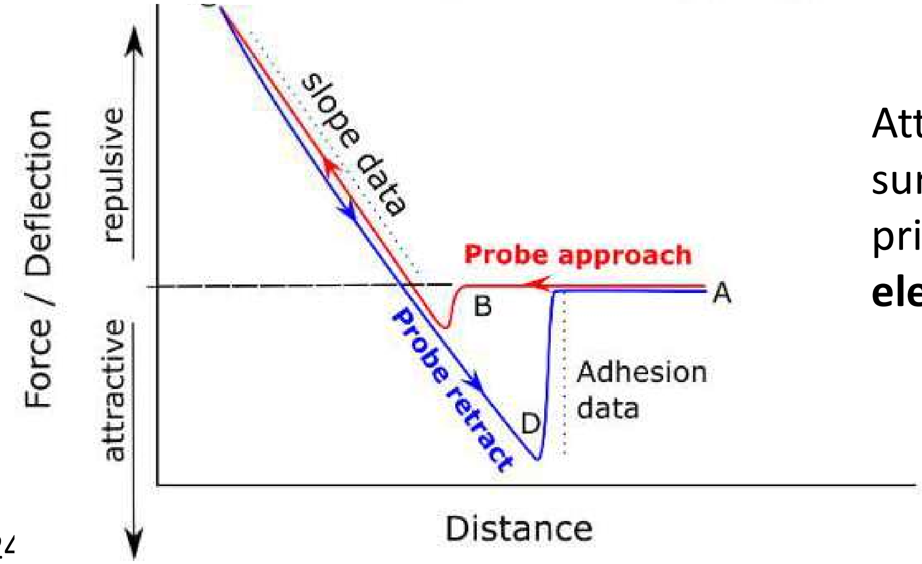

Attractive forces dominate when the tip is far from the surface but still within interaction range. These are primarily **van der Waals forces** and occasionally **electrostatic forces** if the surface or tip is charged.

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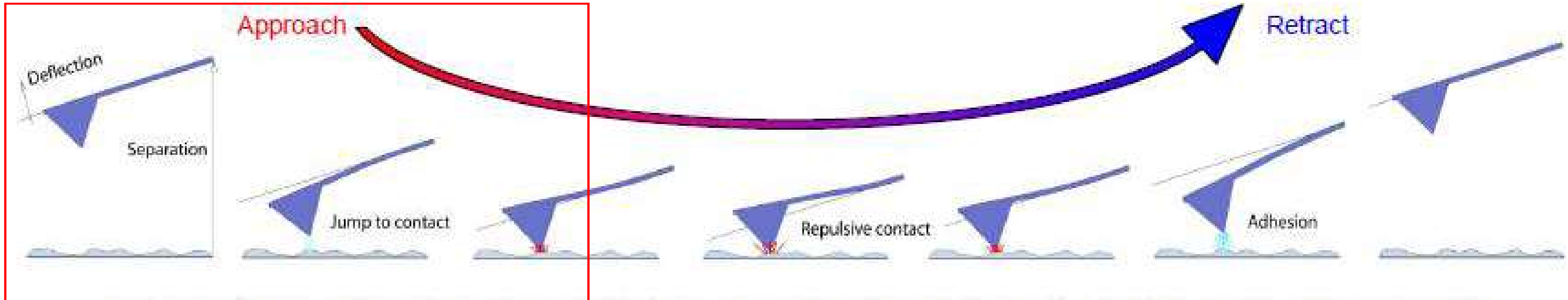
Atomic Force Microscopy

Force –distance curve



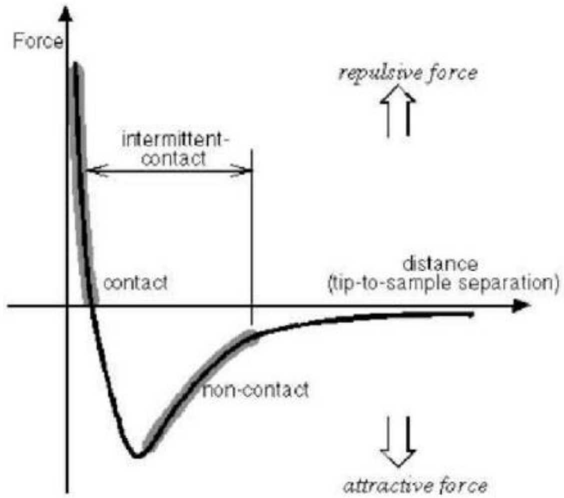
Schematic diagram of the vertical tip movement during the approach and retract parts of a force spectroscopy experiment.

What Are Van der Waals Forces?

Van der Waals (vdW) forces are **weak, attractive intermolecular forces** arising from **electric dipole interactions** between atoms or molecules. Even when a material is electrically neutral, random electron fluctuations create **temporary dipoles**.

These temporary dipoles induce dipoles in nearby atoms → resulting in **attractive force**.

AFM exploits this attraction between the **tip atoms** and **sample atoms**.



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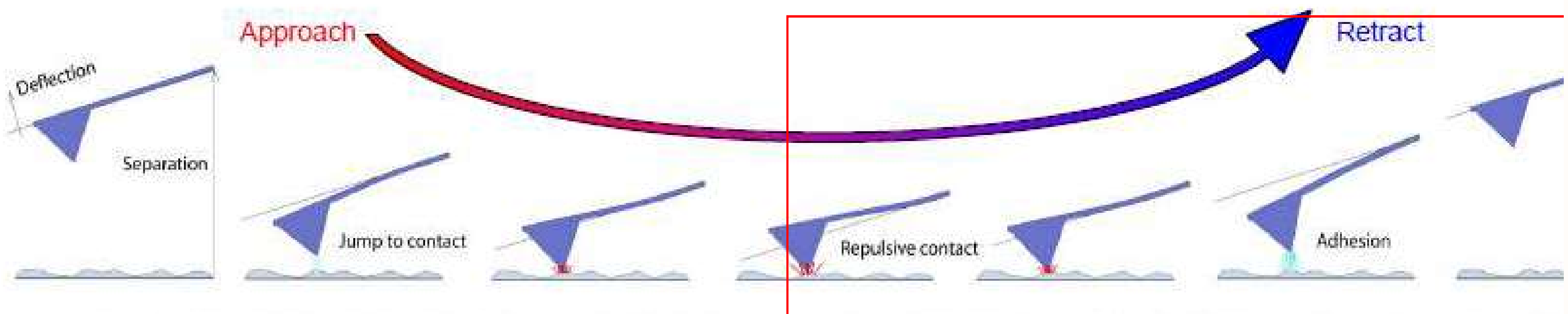
Atomic Force Microscopy Force –distance curve



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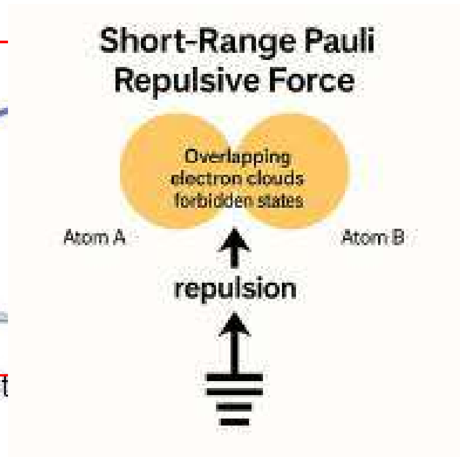
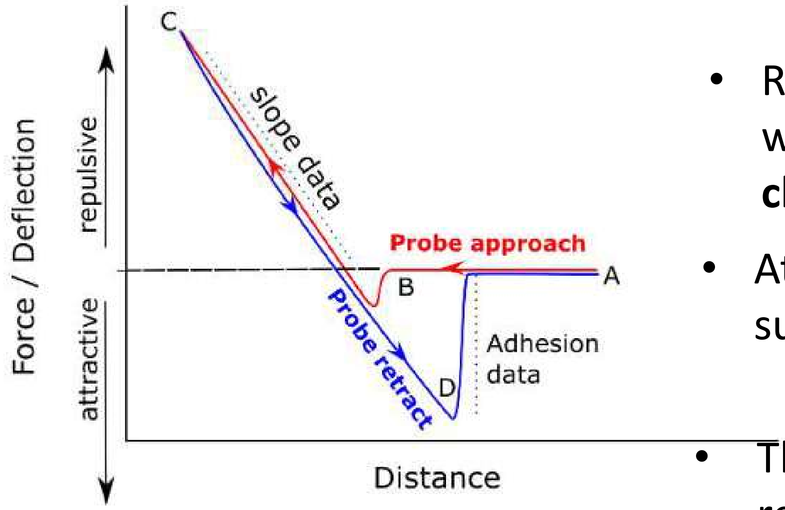




- Repulsive forces dominate when the tip is extremely close to or in contact with the surface. These arise from **Pauli exclusion principle** and **electron cloud overlap**.
- At very short distances, the electron clouds of atoms on the tip and surface start to overlap, creating a strong repulsive interaction.
- The repulsion causes the cantilever to bend away from the surface, resulting in an upward deflection in the graph.

Atomic Force Microscopy

Applications

Adv:

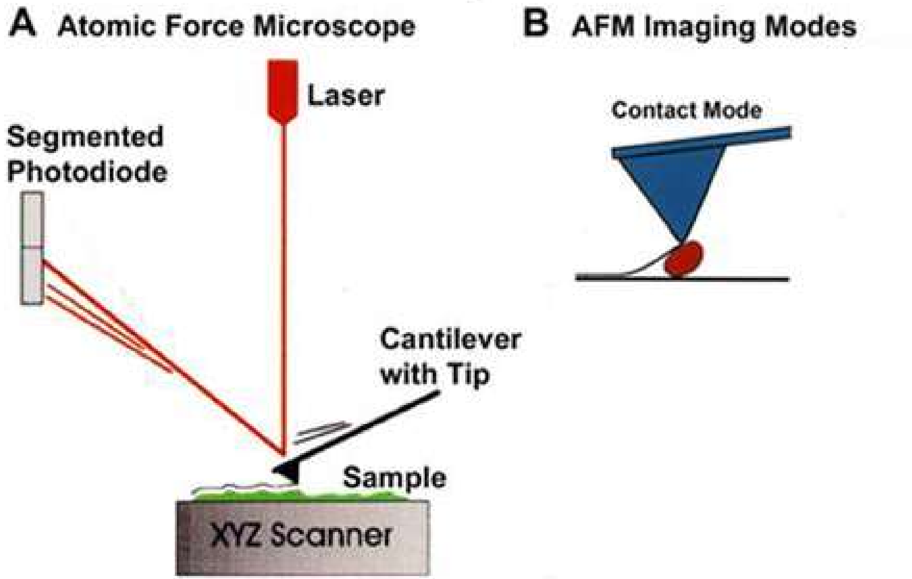
- Easy sample preparation
- Non-destructive imaging
- Accurate height information
- Works in vacuum, air, and liquids
- Sample not required to be conductive
  - Polymers, ceramics, glass
  - Metals
  - Biological samples

Dis-adv:

- Limited vertical range
- Limited magnification range
- Highly Dependent on AFM probes.
- Tip or sample can be damaged

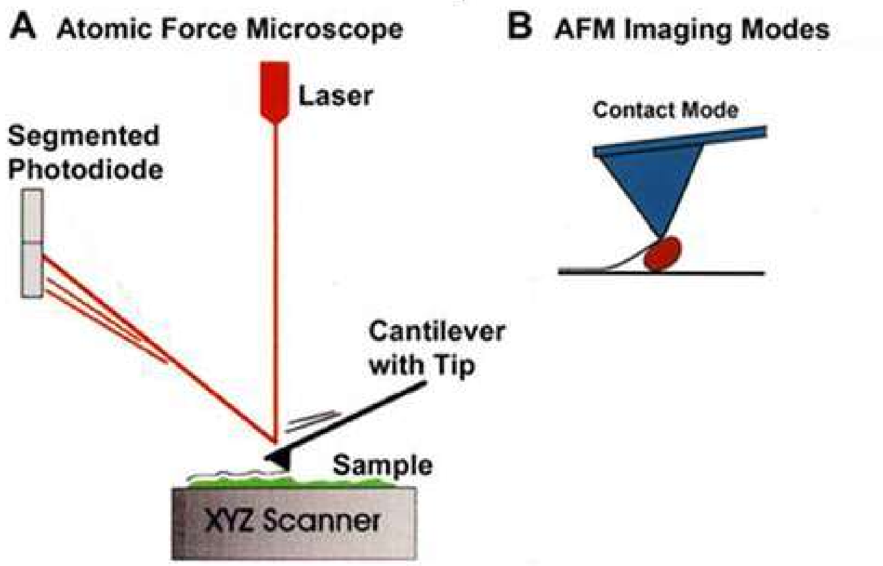
Atomic Force Microscopy

- **Contact AFM**:
  - Contact AFM is the **original and simplest AFM mode**, where the tip maintains **continuous physical contact** with the sample.
  - The forces here are dominated by **repulsive interatomic forces** due to **electron-cloud overlap (Pauli repulsion)**.



Atomic Force Microscopy

• **Contact AFM**:

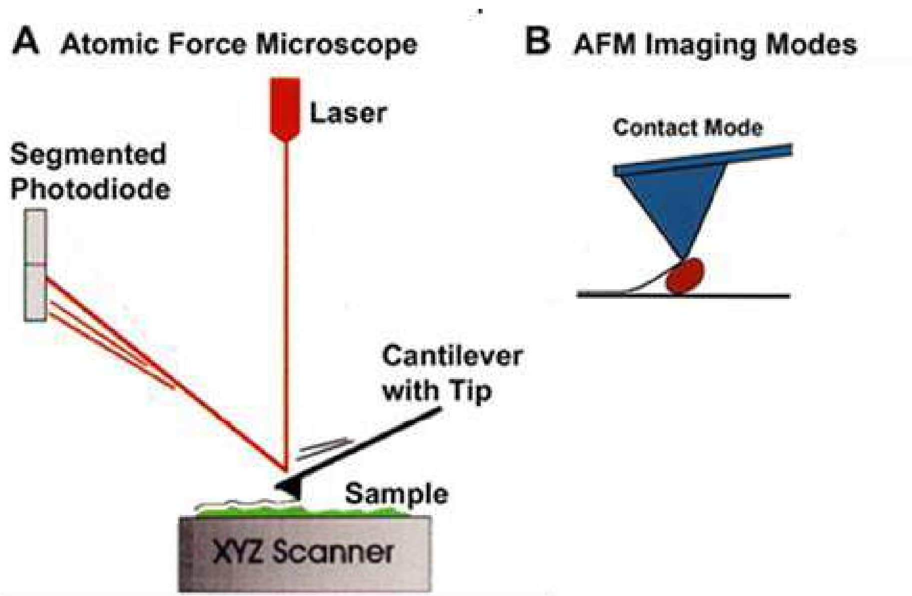


- Contact AFM, or constant force mode, works by dragging a sharp tip attached to a flexible cantilever across a sample's surface while maintaining a constant force.

- When the tip encounters surface features, the cantilever bends. A feedback loop adjusts the vertical position of the sample to keep the force constant, and the recorded movements of the scanner are used to build a high-resolution topographical map of the surface.

Atomic Force Microscopy

- **Contact AFM**:



Tip–Sample Forces in Contact Mode

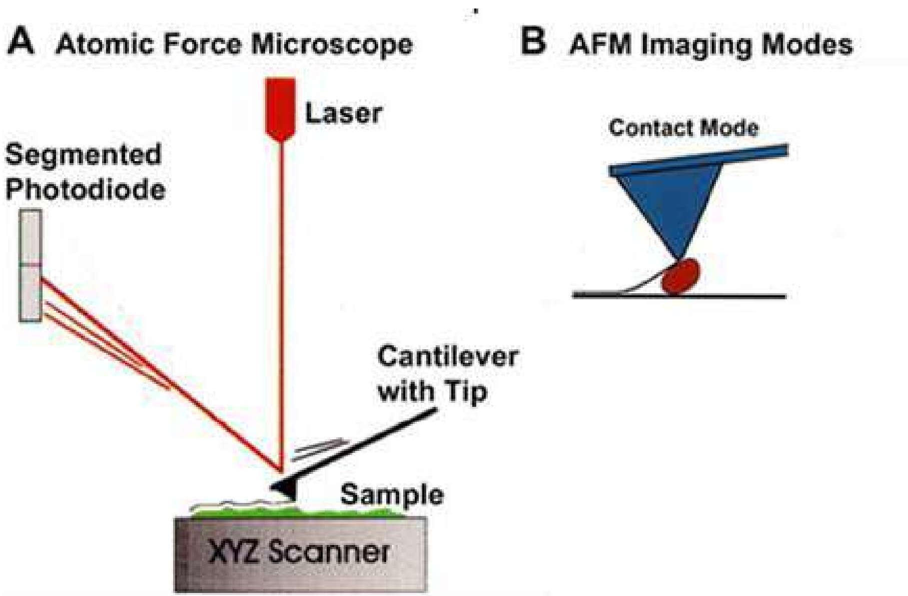
When the tip lightly touches the surface, the interaction is dominated by **short-range repulsive forces**, primarily:

Short-Range Pauli Repulsive Force

- When electron clouds overlap, **electron degeneracy pressure** causes atoms to repel.
- This produces a steep increase in force when tip moves closer than the equilibrium distance.

Atomic Force Microscopy

- **Contact AFM**:



The user sets a **force setpoint** (a specific cantilever deflection).

The feedback loop adjusts the **vertical (Z) piezo** position to keep the force constant.

- If surface rises → tip deflects more → feedback retracts Z
- If surface drops → tip deflects less → feedback extends Z

The resulting Z-motion forms the **height/topography image**.

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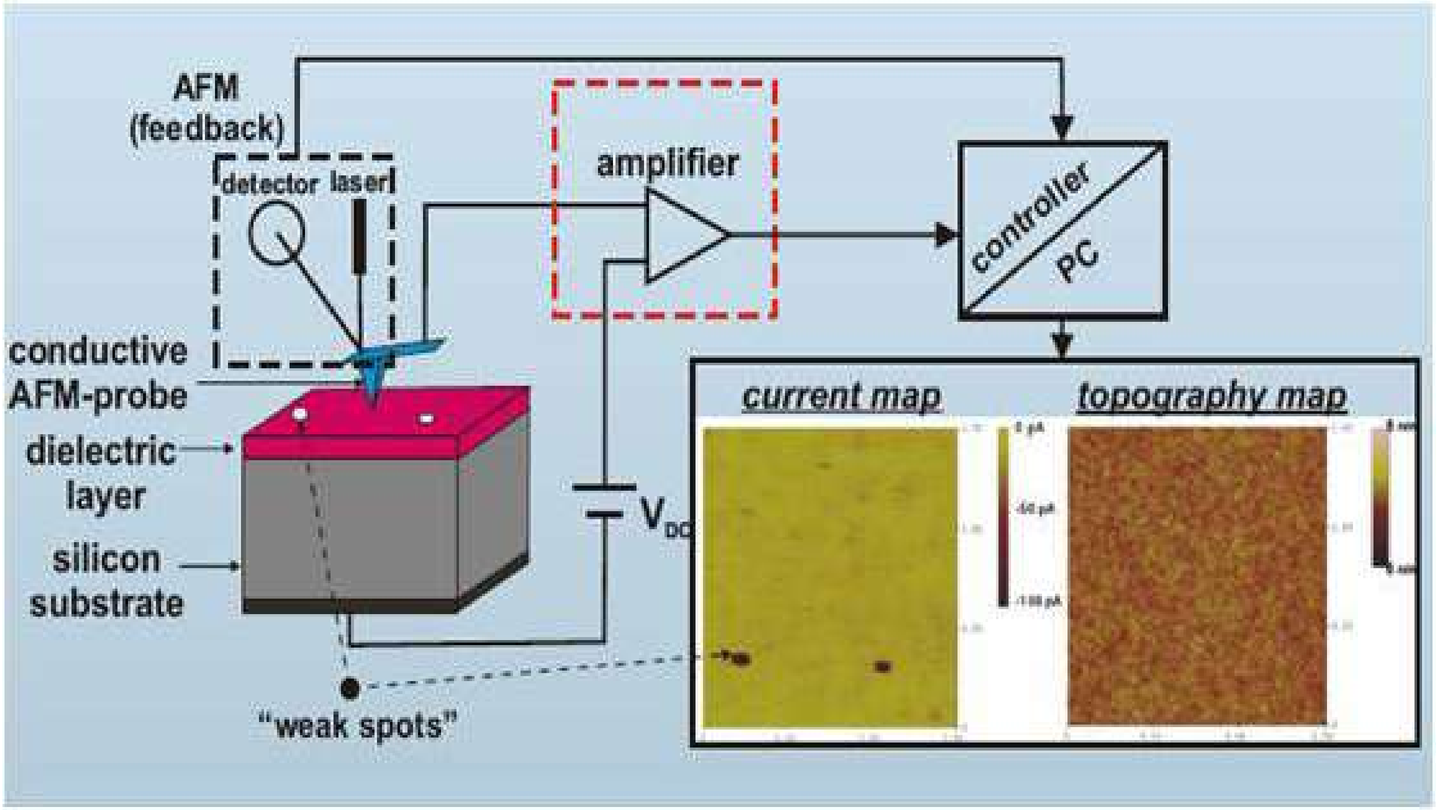
Atomic Force Microscopy

- **Contact AFM**: Topography and current map of thin Dielectric materials (Conductive AFM)- With conductive AFM tip. Ex: Pt-Ir Tip.

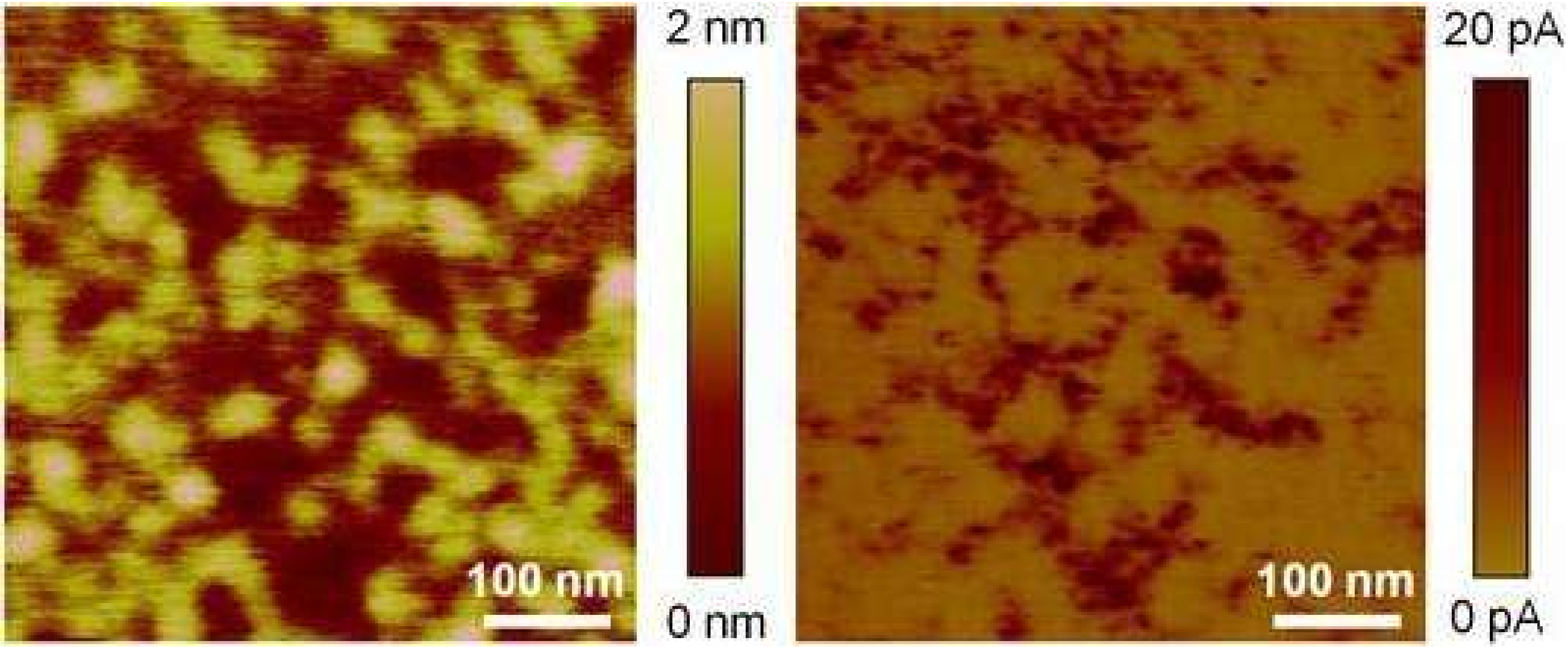


Ref: N. Yao, Microscopy for Nanotech., 2005

Atomic Force Microscopy

Atomic Force Microscopy

- Contact AFM: Topography and current map of thin Dielectric materials (Conductive AFM)



Topographic (left) and current (right) maps collected with CAFM on a polycrystalline HfO_2 stack. The images show very good spatial correlation.

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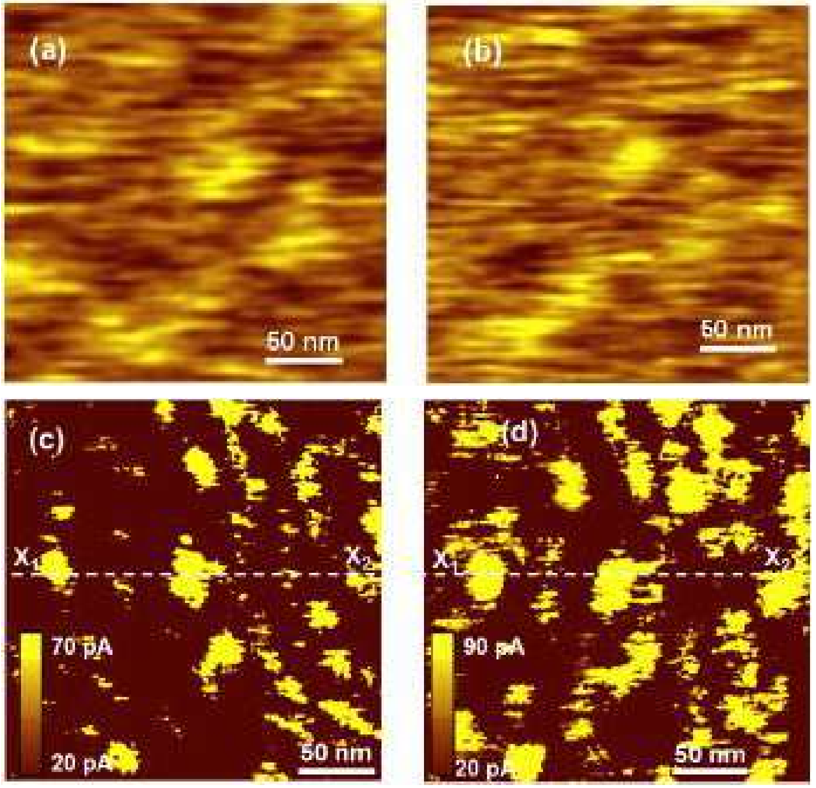
Dilation of leakage spot upon stressing -CAFM

- Larger area scanning: Dilation of leakage spot with Uniform stressing (+6V)

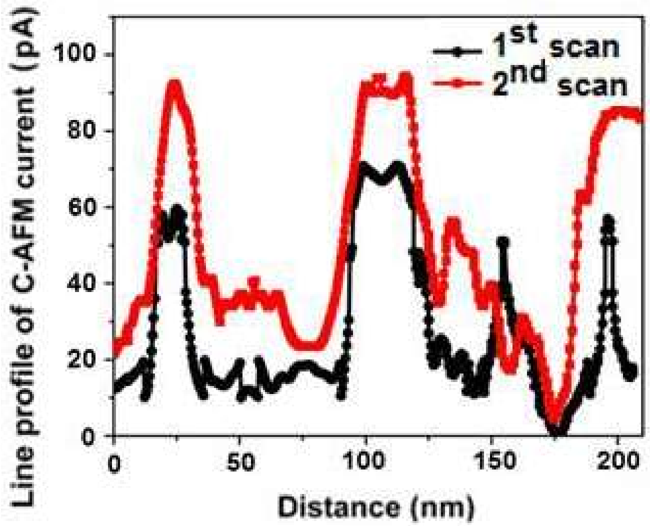
Topography

Current map



First Scan   Second Scan

Along X_1-X_2



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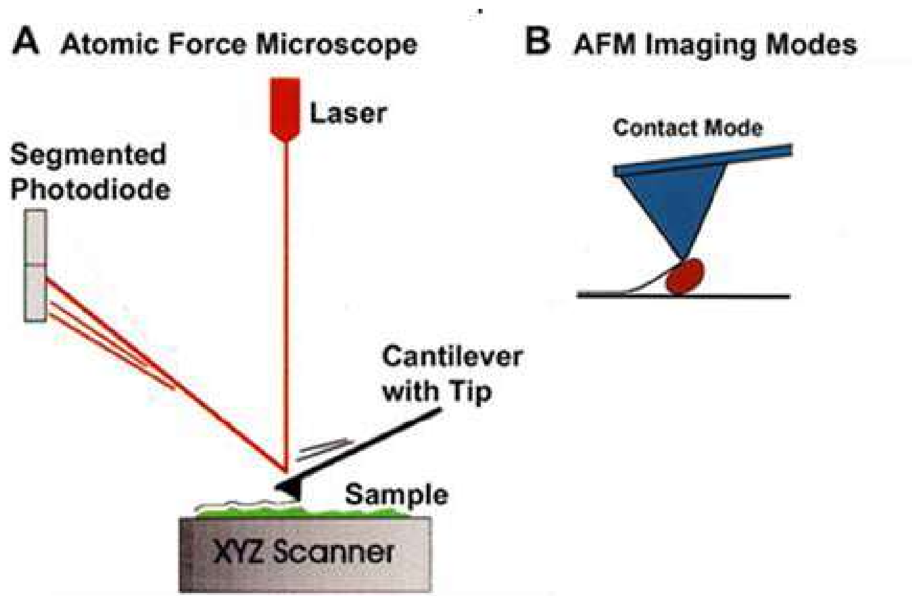
Atomic Force Microscopy

- **Contact AFM**:



Why Contact AFM Works Well

- High imaging speed
- Good for hard materials
- Stable feedback

Issues with Contact Mode

Because the tip drags over the surface:

- Tip wear
- Sample damage (especially polymers, biological samples)

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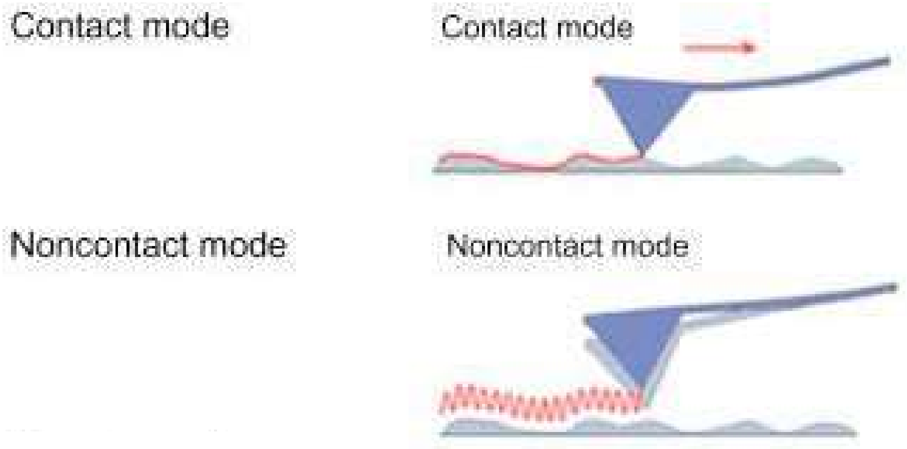
Atomic Force Microscopy

- **Non-contact AFM**: Topography at high resolution for sensitive samples



The two most common modes of nc-AFM operation,

- Frequency modulation (FM)
- Amplitude modulation (AM),

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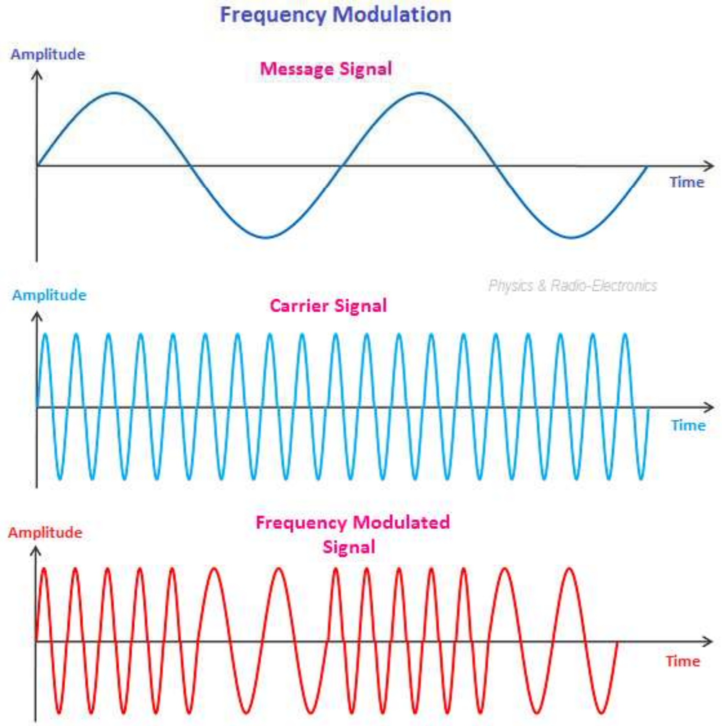
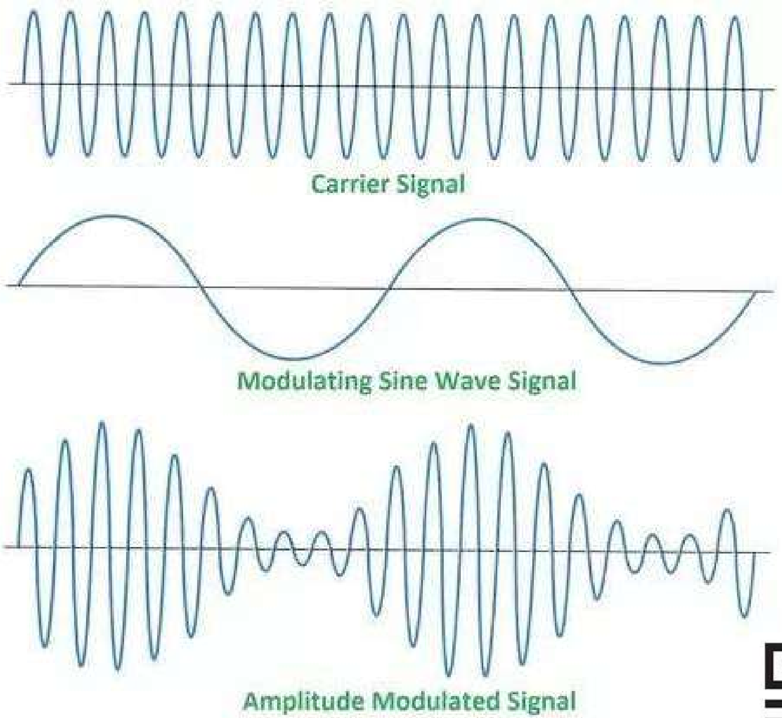
Atomic Force Microscopy

- **Non-contact AFM**:

Frequency modulation (FM)



Amplitude modulation (AM)



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Atomic Force Microscopy

- **Non-contact AFM**:
  - oscillating cantilever, tip not in contact with sample
  - used for soft samples
  - imaging in vacuum

Non-contact imaging (center) generally provides low resolution and can also be hampered by the contaminant (e.g., water) layer which can interfere with oscillation.

Atomic Force Microscopy

- **Non-contact AFM**:

Non-Contact AFM (NC-AFM) – Working Principle

Non-contact AFM operates with the cantilever tip **oscillating near the sample surface without touching it**.

The tip is kept at a distance where only **long-range attractive forces** act — mostly **van der Waals**, and **electrostatic**,forces.

This mode avoids tip wear and sample damage, making it ideal for **soft materials, 2D materials, and atomic-scale imaging** (especially in UHV).

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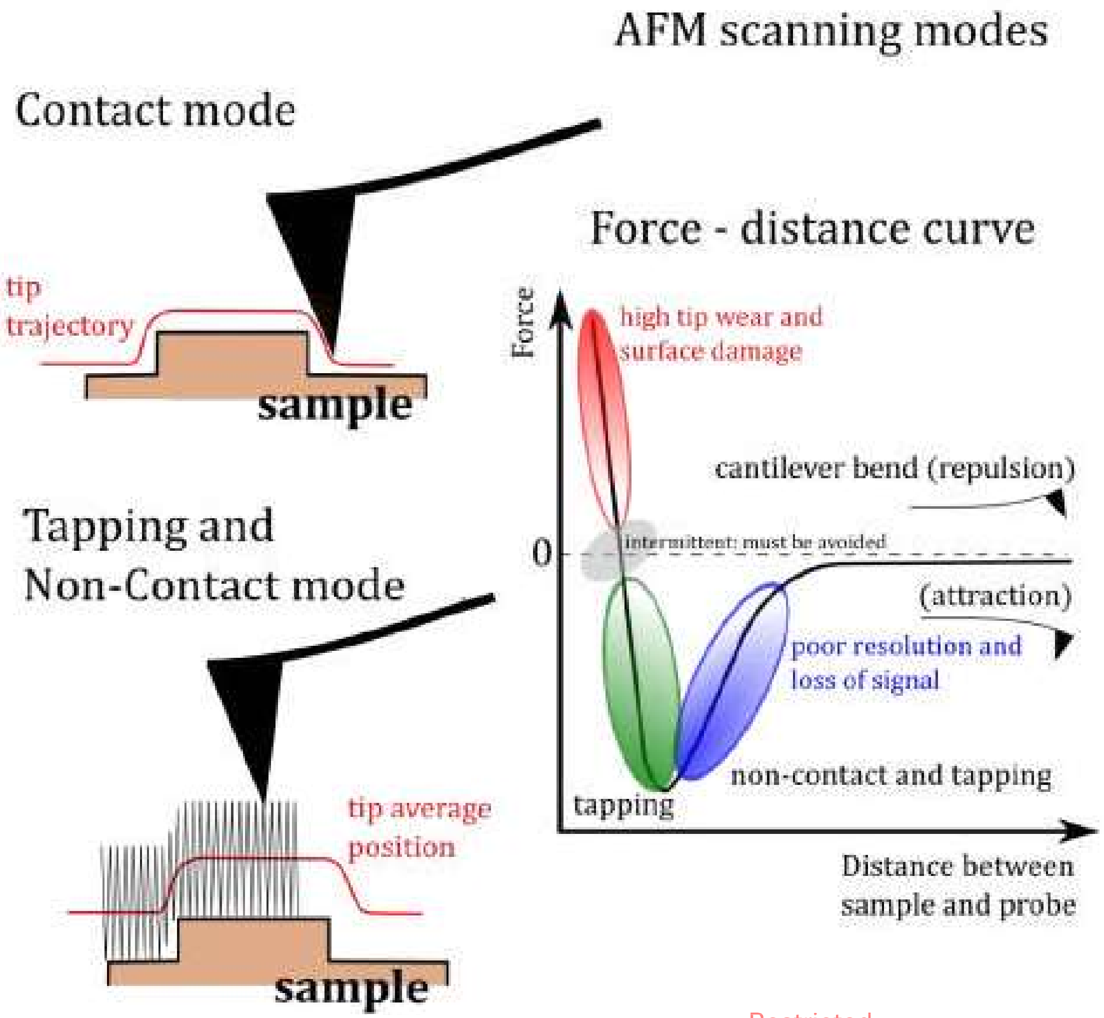
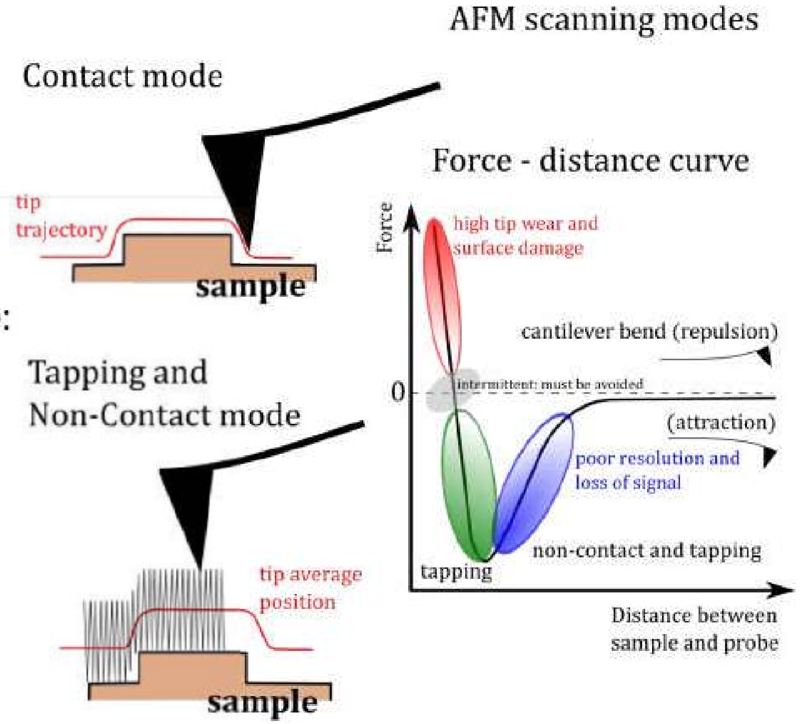
Atomic Force Microscopy

- **Non-contact AFM**:



Atomic Force Microscopy

• **Non-contact AFM: Principle**



Forces Involved (Tip–Sample Interaction)

In NC-AFM, the tip is kept in the **attractive region** of the force–distance curve:

Dominant Forces in Non-Contact Mode

- van der Waals force
- Electrostatic forces

No Pauli repulsion

The tip is kept far enough away so that **electron cloud overlap does not occur**, hence **no mechanical contact, negligible tip wear**.

Atomic Force Microscopy • Non-contact AFM: Principle

How Does NC-AFM Detect the Surface?

When the tip approaches the sample, **attractive forces cause a slight change in the effective spring constant** of the cantilever:

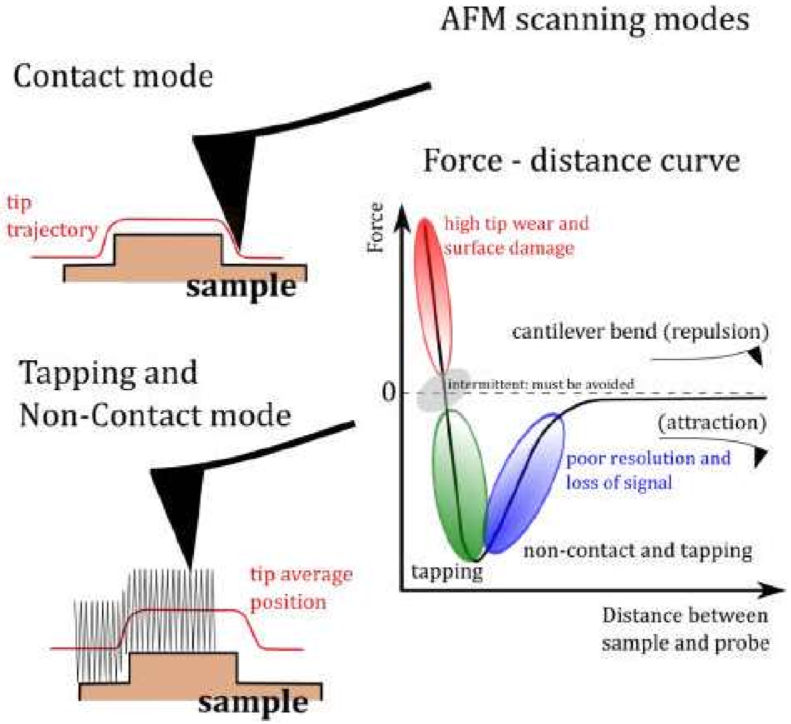
$$k_{\text{eff}} = k - \frac{\partial F}{\partial z}$$

This modifies the **resonance frequency:**

$$f = \frac{1}{2\pi}\sqrt{\frac{k_{\text{eff}}}{m}}$$

So, the surface is detected by measuring:

- Frequency shift Δf
- Phase shift $\Delta\varphi$
- Amplitude change



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Atomic Force Microscopy • **Non-contact AFM: Principle**

The cantilever is driven at resonance:

$$f_0 = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$

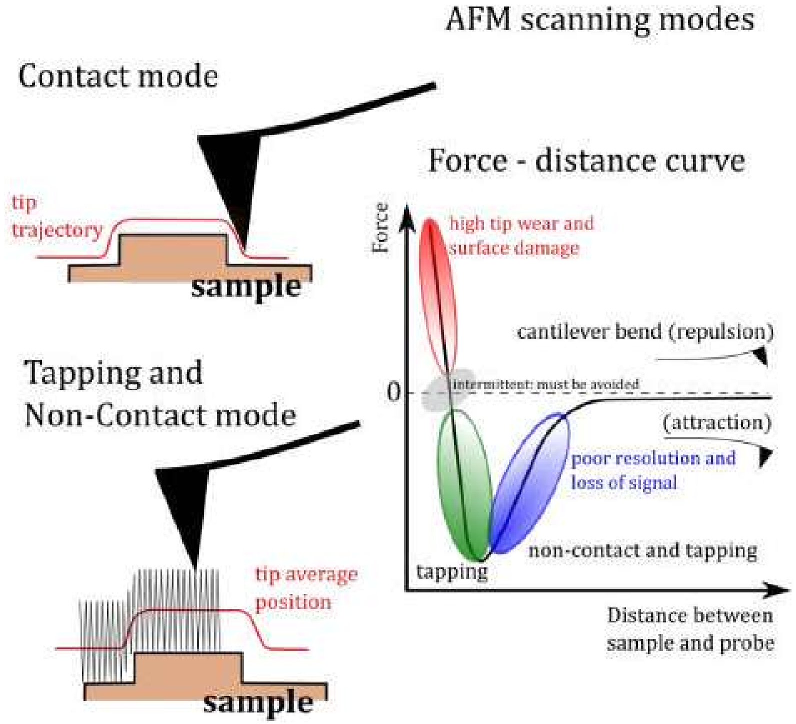
Its oscillation amplitude A_0 remains constant under feedback control.

As the oscillating tip nears the sample:

- **Attractive van der Waals force increases**
- **Effective spring constant decreases**
- **Resonance frequency drops**

This produces a **negative frequency shift:**

$$\Delta f = f - f_0 < 0$$



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Atomic Force Microscopy

• Non-contact AFM: Principle

Feedback adjusts the **Z-piezo height** to keep Δf constant:

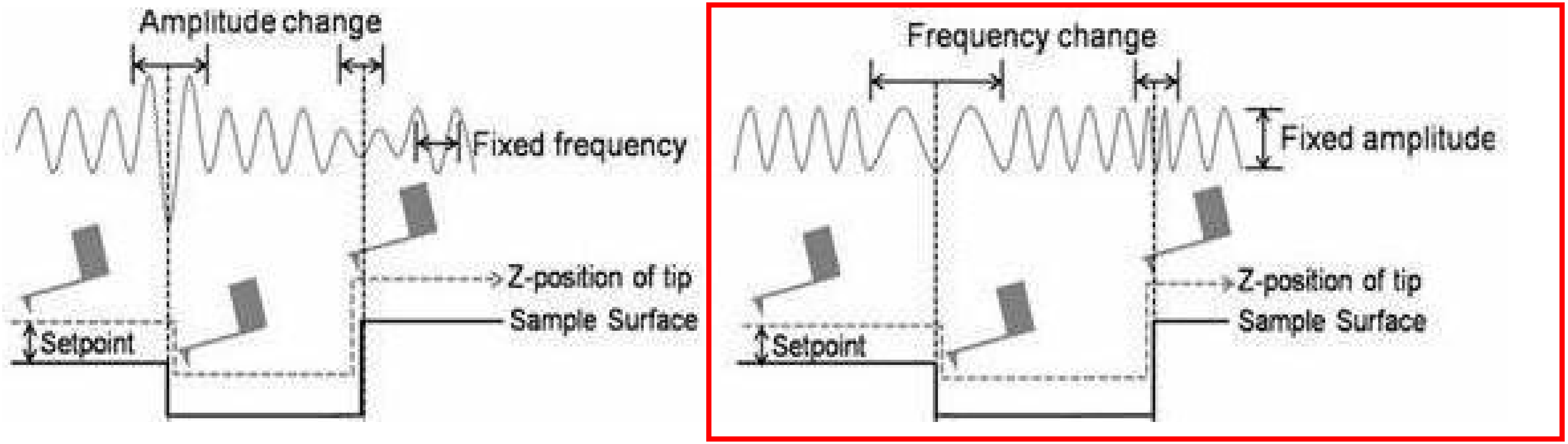
- If the surface rises → attractive force increases → Δf becomes more negative → feedback retracts tip
- If the surface dips → attractive force decreases → feedback lowers tip

Thus, the **Z signal forms the topography image.**

Atomic Force Microscopy

- **Non-contact AFM**:

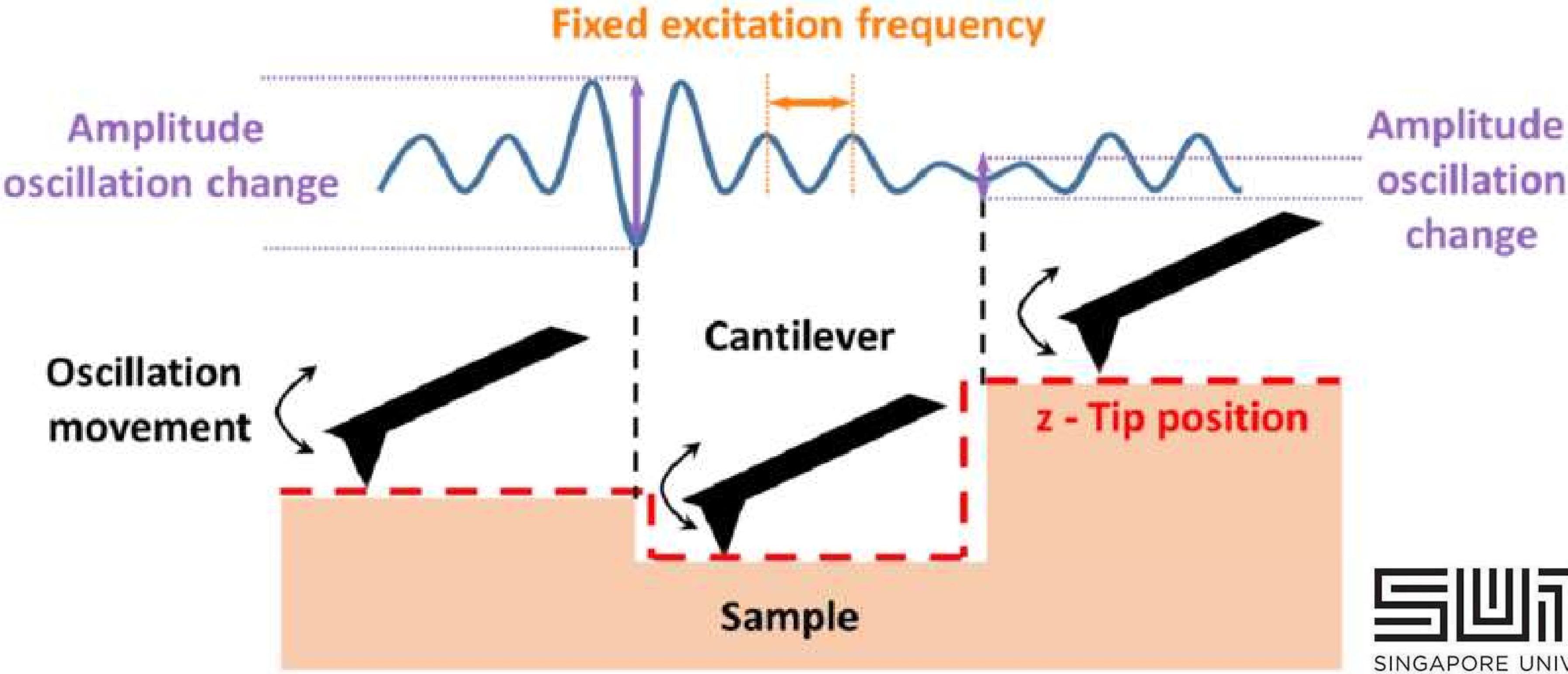
Frequency modulation (FM)



Atomic Force Microscopy

- **Non-contact AFM**:

Amplitude modulation (AM)



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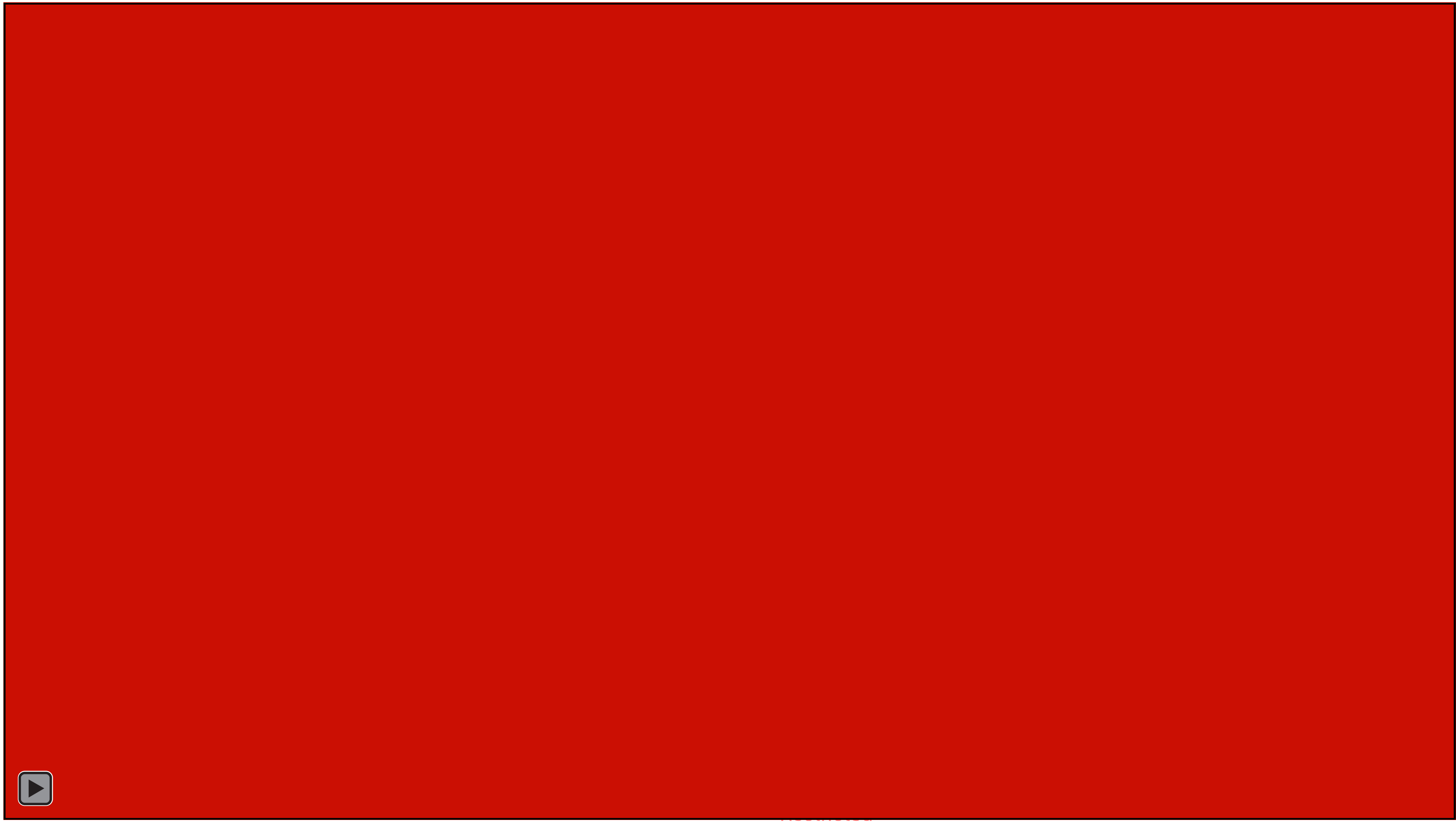
Atomic Force Microscopy



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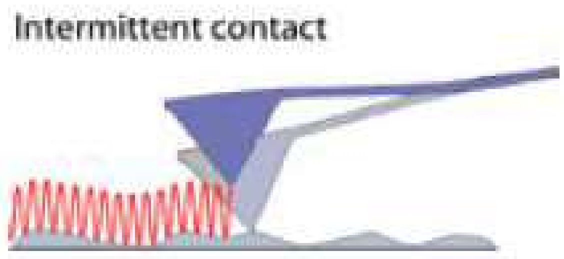
Atomic Force Microscopy

- **Tapping Mode AFM**:

  - oscillating cantilever, tip touching surface gently and frequently
  - often used for biological samples
  - imaging in air and liquid
  - good resolution



Atomic Force Microscopy

AFM Modes of Operation	Working Principle	Advantage	Disadvantage
Contact Mode	• Physical contact between the tip and the surface	• High scan speeds	• Damage to soft sample • Later forces may produce image artefacts
Non-contact Mode	• No contact between the tip and the sample	• High resolution • No damage to sample	• Slower scan speed if compared with both contact and tapping mode
Tapping Mode	• Intermittent and short contact between the sample and the tip	• Minimal damage to sample	• Slower scan speed if compared with contact mode